S_{plan}: Compact Learned Symbolic Planning Tokens for Music-Aware Audio Language Models

Dabin Kim* **Minhee Lee*** **Ji Jiaxian*** **Aner Zheng***
Graduate School of Cultural Technology, KAIST
{dabinkim, minheelee, jiaxian, anerzheng}@kaist.ac.kr

Abstract

We ask three questions about symbolic music representations in audio language models: (1) does a compact learned symbolic stream, aligned to an audio window's own MIDI, reduce acoustic token prediction uncertainty more than a shuffled or dummy stream of the same size? (2) is that benefit specific to content alignment, or merely to token capacity? (3) does the symbolic stream encode music-structural attributes recoverable by probing?

We address these with S_{plan}, a compact symbolic planning token that compresses MIDI-derived musical structure into a fixed-length residual vector quantization (RVQ) stream inserted between the reasoning and acoustic token streams of UniAudio 2.0. On questions (1) and (2): across three independent controlled comparisons using aligned, shuffled, and dummy conditions, aligned S_{plan} consistently reduces C_a cross-entropy below both corrupted controls, most rigorously in our final hardening run (test CE 5.600 vs. 5.614 for shuffled and dummy, 5.859 for reason-only), confirming an alignment-specific effect of $\Delta = -0.014$ on top of a capacity effect of $\Delta = -0.245$. On question (3): MIR probing recovers weak key accuracy 0.557 ($+0.431$ over the zero-feature baseline) and pitch-class macro-F1 0.854 ($+0.024$), while meter and instrument micro-F1 are prior-dominated and excluded from claims. Two negative results (side-channel insertion and adapter gating) establish a stream-native interface as a necessary design constraint. The central claim is conservative but supported: S_{plan} is a compact symbolic codec with acoustic-token utility and recoverable music-structural content under controlled Slakh experiments.

1 Introduction

Unified audio language models (LALMs) (**????**) predict discrete audio tokens autoregressively and have achieved strong performance across diverse tasks. Music, however, poses a persistent gap: the structural properties most salient to musicians and listeners—harmonic progressions, rhythmic regularity, instrument co-occurrence, phrase boundaries—are encoded in MIDI symbolic representations, not waveform statistics. While MIDI-to-audio synthesis pipelines use explicit symbolic intermediates, integrating those representations into LALMs in a *controlled, verifiable* way has remained open.

What we are trying to prove. We organize our work around three research questions (RQs):

RQ1 (Acoustic Utility): Does a compact learned symbolic stream, aligned to an audio window's own MIDI, reduce acoustic token prediction uncertainty more than a shuffled or dummy stream of the same length?

RQ2 (Alignment Specificity): Is the benefit from alignment-specific content, or from token capacity alone? We test this by requiring aligned to beat *both* shuffled (content broken, statistics preserved) and dummy (zero content, same length) controls under matched training conditions.

RQ3 (Music-Structural Content): Does the symbolic stream encode musically meaningful attributes recoverable by supervised probing? We test pitch, key, meter, density, and instrument recognition, each compared against a zero-feature baseline that exposes dataset priors.

Answering RQ1 affirmatively is not automatic: language models are good at ignoring spurious context. We demonstrate exactly this failure mode when the symbolic tokens are inserted as an external side-channel incompatible with the model's positional encoding — aligned and shuffled MIDI insertions then produce nearly identical predictions. Our positive result for RQ1 is therefore contingent on the correct interface design established through two preliminary negative results.

Answering RQ2 is important because any fixed-length stream adds context capacity. A positive result on RQ1 alone is weak evidence if an equal-length dummy stream achieves the same gain. Our multi-condition design forces the claim: aligned must beat both shuffled and dummy, or we cannot attribute the gain to symbolic content.

Answering RQ3 connects acoustic utility to an interpretable mechanism. A symbolic stream that reduces cross-entropy but encodes nothing auditable could be acting through any latent channel; MIR probing makes the representational content explicit and falsifiable.

How we answer them. We introduce S_{plan}, a *learned symbolic planning token* that compresses MIDI-derived features into a compact, fixed-length, multi-stream RVQ representation and inserts it between the reasoning tokens R_a and semantic acoustic tokens C_a of UniAudio 2.0 (**?**), forming $R_a \to S_{\text{plan}} \to C_a$. We evaluate through a ladder of controlled comparisons with aligned / shuffled / dummy conditions at each stage, plus MIR probing with a zero-feature baseline.

Contributions.

1. A low-rate RVQ symbolic codec (S_{plan}) that compresses MIDI-derived frame features into 4×30 tokens per 30-second window—a $10\times$–$50\times$ reduction versus raw MIDI tokenizations.

2. **A positive answer to RQ1:** aligned S_{plan} reduces C_a cross-entropy consistently across three independent controlled comparisons at different training scales.

3. **A positive answer to RQ2:** aligned beats *both* shuffled and dummy in all three comparisons; the alignment-specific effect is $\Delta \in [-0.034, -0.014]$ CE and is consistent in sign across independent training runs.

4. **A qualified answer to RQ3:** MIR probing recovers weak key and pitch-class attributes well above the zero-feature baseline, while meter and instrument micro-F1 are excluded as prior-dominated.

5. Two negative results that establish stream-native interface as a necessary design constraint, not merely an engineering preference.

2 Related Work

2.1 Audio Language Models

AudioLM (**?**) pioneered hierarchical discrete token prediction for long-form audio continuation. AudioPaLM (**?**) unified speech and audio tasks under a single LLM. UniAudio (**?**) extended this to a broad task catalog using a multi-stream tokenizer. UniAudio 2.0 (**?**) introduced ReasoningCodec, which factorizes audio into reasoning tokens (high-level, language-aligned) and reconstruction tokens (fine-grained acoustic detail), supervised via SFT and GRPO. UALM (**?**) uses a Qwen2.5-based decoder with continuous audio embeddings and X-codec tokens, demonstrating competitive MMAU performance but without explicit symbolic grounding.

None of these systems incorporate a learned *symbolic* planning stream verified by alignment controls. Our work fills this gap on the music domain.

2.2 Symbolic Music Representations

Symbolic music can be represented via raw MIDI events, piano rolls, or higher-level abstractions (**??**). Raw MIDI tokenizations suffer from extreme sequence lengths and are not directly comparable to audio token rates. Compact representations such as chord sequences or beat annotations are hand-designed and limited in expressiveness. S_{plan} learns a compact representation end-to-end from paired MIDI features, avoiding hand-design while keeping sequence length tractable.

2.3 Vector Quantization for Audio

Residual vector quantization (RVQ) has become the standard for neural audio codecs (**???**). We apply the same quantization principle to symbolic features rather than waveforms, enabling S_{plan} to use the same multi-stream position encoding as the acoustic codec.

2.4 Music Information Retrieval

MIR tasks including key estimation, beat tracking, instrument recognition, and chord detection provide objective, structured evaluation targets (**?**). We use MIR attributes from Slakh MIDI metadata as supervision and evaluation targets for S_{plan}, yielding interpretable quality measures beyond cross-entropy.

3 Background: UniAudio 2.0

UniAudio 2.0 (?) organizes audio into two factorized streams:

Reasoning tokens R_a: Low-rate, language-aligned tokens from a ReasoningCodec encoder, trained with SFT + GRPO on ASR, captioning, and audio-reasoning tasks. These tokens capture high-level semantic and contextual content.

Reconstruction tokens C_a: An 8-codebook RVQ stream for fine-grained acoustic reconstruction and generation, trained after the reasoning branch is frozen.

The unified autoregressive model predicts C_a conditioned on text and R_a, using a single transformer with multi-stream positional encoding. Audio-specific experts update only audio-position hidden states. This factorization allows both understanding (via R_a) and generation (via C_a) within a single framework.

Our work asks: *can a symbolic planning stream S_{plan}, inserted between R_a and C_a, carry music-structural information that R_a alone does not provide?*

4 The S_{plan} System

4.1 MIDI Feature Extraction

Given a 30-second Slakh audio-MIDI window, we extract frame-level features from the aligned MIDI file at 100 ms resolution:

- **Binary features:** Instrument-class presence vector (34 Slakh classes), 12-dim pitch-class binary, beat/downbeat phase indicator.
- **Continuous features:** Instantaneous tempo (BPM), note density (events/second), polyphony count, instrument-weighted energy proxy.

Features are normalized per-dimension across the training split. The final frame vector has 53 dimensions (41 binary, 12 continuous).

4.2 Symbolic RVQ Codec

We train a residual vector quantizer over the frame-level feature sequence:

- **Encoder:** Two-layer 1D CNN followed by temporal pooling that maps variable-length feature sequences to a fixed 30-frame representation per 30-second window.
- **Codebook:** 4 RVQ stages, 512 codes each, trained with exponential moving average updates and codebook restart.
- **Decoder:** Symmetric 1D CNN that reconstructs the original feature sequence from quantized codes.
- **Loss:** Binary cross-entropy for binary feature dimensions plus mean squared error (MSE) for continuous dimensions.

Training runs for 42,000 steps on the Slakh training split.

4.3 Stream-Native Integration

We integrate S_{plan} into the UniAudio 2.0 sequence as a third stream:

$$\underbrace{[\text{text}]}_{\text{1 stream}} \rightarrow \underbrace{[R_a]}_{\text{8 streams}} \rightarrow \underbrace{[S_{\text{plan}}]}_{\text{4 streams}} \rightarrow \underbrace{[C_a]}_{\text{8 streams}}$$

The layout produces 151 reason frames, 30 symbolic frames, and 376 semantic frames (total 557 positions), fitting within the 1024 context limit with 0% truncation.

We use LoRA adapters (?) on the UniAudio 2.0 transformer attention modules for parameter-efficient fine-tuning (140 modules, $\sim$10.8M trainable parameters in the reason-only configuration and $\sim$253M for symbolic stream variants), with a learned scalar gate that controls symbolic stream influence magnitude.

4.4 Why Not Raw MIDI?

Our design is motivated by three failures of raw MIDI appending:

- **Length:** Standard MIDITok (?) tokenizations of 30-second windows average $\sim$2,000–8,000 tokens, saturating the 1024 context.
- **Alignment:** Raw MIDI event timestamps are asynchronous with audio frames, creating positional artifacts incompatible with multi-stream encoding.

- **Controllability:** Variable-length event sequences do not permit fixed-length shuffled/dummy controls, making the evaluation design impossible.

Low-rate rule summaries (hand-crafted text-like symbolic summaries, $\sim$149 tokens) avoid the length problem but remain hand-designed. §6.2 shows that learned S_{plan} outperforms rule summaries in acoustic-token utility.

5 Experimental Setup

5.1 Dataset

All experiments use Slakh2100 (**?**), a multi-track paired audio-MIDI dataset synthesized with professional-grade sample-based virtual instruments. We split at the *track* level before windowing to prevent train-test contamination:

Split	Tracks	30 s windows	Notes
Train	1,289	11,279	Track-level split
Validation	270	2,334	Zero track overlap
Test	151	1,385	Held-out tracks

Table 1: Slakh dataset statistics. All comparisons use the same split.

5.2 Control Protocol

For every controlled comparison, we train and evaluate three symbolic conditions under matched hyperparameters:

Aligned S_{plan}: Tokens from the audio window's own MIDI file, preserving temporal and content alignment.

Shuffled S_{plan}: Tokens of a randomly selected *different* window, preserving token statistics but breaking content alignment.

Dummy stream: Zero or uniformly random RVQ codes, providing a token-budget control without any symbolic content.

A gain of aligned over *both* shuffled and dummy is required to attribute any improvement to alignment-specific content (RQ2).

5.3 Metrics

- **Cross-entropy (CE) / perplexity (PPL):** Token-level prediction loss on C_a acoustic tokens.
- **Token accuracy:** Exact-match accuracy on held-out C_a tokens.
- **Symbolic gate:** The learned gate value, measuring the fraction of symbolic stream influence admitted to the acoustic prediction path.
- **MIR metrics:** Instrument macro/micro-F1, pitch-class macro-F1, meter accuracy, weak key accuracy, segment density accuracy, and segment polyphony accuracy.

6 Results

We present results organized around the three research questions. §6.1 establishes that the symbolic codec is technically sound — a necessary precondition for the RQ1–RQ3 claims. §6.2 and §6.3 jointly address RQ1 and RQ2 through three controlled comparisons at increasing training scale. §6.4 addresses RQ3 via MIR attribute probing. §6.5 documents two negative results that establish stream-native interface as a necessary design constraint.

6.1 Prerequisite: Codec Reconstruction Quality

Before asking whether S_{plan} carries useful symbolic information downstream, we verify that the RVQ codec faithfully reconstructs the input symbolic features. We train the codec for 42,000 steps on the Slakh training split and evaluate on the validation set.

Metric	Value
Reconstr. loss (val.)	0.00918
Binary feature acc.	0.99777
Continuous MSE	0.00172
Codebook util. bk1	31%
Codebook util. bk2	72%
Codebook util. bk3	81%
Codebook util. bk4	81%

Table 2: Codec reconstruction quality. Per-window binary accuracy is 1.000 on all three validation fixed examples. Books 2–4 reach 72–81% utilization: no codebook collapse.

The codec produces a technically healthy symbolic bottleneck: it compresses MIDI-derived structure without collapse and reconstructs features faithfully. This is a prerequisite, not a claim — it establishes that differences in downstream experiments are attributable to symbolic content, not codec failure.

6.2 RQ1: Aligned S_{plan} Reduces Acoustic Token Uncertainty

We test RQ1 through three controlled comparisons in order of increasing rigor.

Proxy prediction task. We first train a small transformer head to predict C_a tokens from frozen R_a + symbolic representations, treating this as an independent proxy of acoustic utility (5,000 steps, 1,385 test windows).

Condition	Valid CE	Test CE	Test PPL
Reason only	7.5650	7.6573	2116.1
Low-rate rule summary	7.5534	7.6445	2089.2
Aligned S_{plan}	**7.5073**	**7.5963**	**1990.9**
$\hookrightarrow$ Shuffled S_{plan}	7.5376	7.6307	2060.6
$\hookrightarrow$ Dummy stream	7.6023	7.6942	2195.5
$\hookrightarrow$ Random stream	7.5400	7.6325	2064.2

Table 3: Proxy task comparison on 1,385 held-out test windows. Aligned S_{plan} achieves the lowest test CE across all conditions including corrupted controls.

Aligned S_{plan} achieves the lowest test CE (7.5963), outperforming reason-only (-0.061), low-rate rule summaries (-0.048), shuffled (-0.034), and dummy (-0.098). Learned S_{plan} outperforms the hand-crafted rule summary, confirming the value of end-to-end symbolic compression. This is the first evidence that S_{plan} carries acoustically useful information beyond what R_a already encodes.

Stream-native warmup. We next implement the full three-stream layout $R_a \to S_{\text{plan}} \to C_a$ with streams aligned to the base model's positional encoding, using a shorter warmup budget to validate the interface design (3,000 steps).

Condition	Test CE	Test Acc	Gate
Aligned S_{plan}	**5.766**	**0.1097**	0.0033
$\hookrightarrow$ Shuffled S_{plan}	5.785	0.1067	0.0032
$\hookrightarrow$ Dummy stream	5.785	0.1068	0.0034

Table 4: Stream-native comparison (sequence length 557, 0% truncation). All three conditions differ only in the symbolic stream variant.

Aligned achieves test CE 5.766 vs. shuffled 5.785 ($\Delta = -0.018$) and dummy 5.785 ($\Delta = -0.019$). The margin is modest but consistent; on four validation fixed windows, two show strong positive margins (+0.054, +0.004) and two show near-zero or slightly negative margins (-0.006, -0.007), characteristic of a real but small signal that has not saturated the learning budget.

Complete four-condition hardening (primary result). Our most rigorous evaluation runs all four conditions under identical hyperparameters (800 steps, batch 1, gradient accumulation 16, LoRA lr 10^{-5}, symbolic lr 3×10^{-4}, gate lr 2×10^{-3}, seed fixed), differing only in the symbolic stream variant.

Condition	Test CE	Test Acc	Gate
Reason-only baseline	5.859	0.1039	0.000
Aligned S_{plan}	**5.600**	**0.1207**	0.177
$\hookrightarrow$ Shuffled	5.614	0.1191	0.167
$\hookrightarrow$ Dummy	5.614	0.1193	0.163

Table 5: Complete four-condition comparison. All conditions trained identically; only the symbolic stream variant differs. Primary empirical result.

Aligned achieves test CE 5.600, substantially below reason-only (-0.259) and below both corrupted controls (-0.014). These results answer **RQ1 positively**: aligned S_{plan} consistently reduces acoustic token uncertainty across three independent controlled comparisons at different training scales.

6.3 RQ2: The Benefit Is Alignment-Specific, Not Token Capacity

Any fixed-length stream added to R_a expands the context available for C_a prediction. RQ2 asks whether the gains in §6.2 are attributable to this capacity effect, or to the aligned content specifically.

Decomposing the gain. The $\Delta = -0.259$ CE improvement of aligned over reason-only in our primary evaluation (Table 5) decomposes as:

- **Stream capacity effect:** Any symbolic stream variant outperforms reason-only by $\Delta \approx -0.245$ CE (PPL: 350 $\to$ 274), reflecting the additional context length regardless of content.
- **Alignment-specific effect:** Aligned S_{plan} outperforms both shuffled ($\Delta = -0.014$) and dummy ($\Delta = -0.015$). The shuffled condition preserves token statistics while breaking temporal alignment; the dummy condition removes all symbolic content. Aligned beating *both* isolates content alignment as the source of the additional gain.

Consistency across training scales. The alignment-specific margin decreases from the proxy task to the hardening run, likely because the

Comparison	Steps	Δ (aligned vs. shuffled)
Proxy task	5,000	-0.034
Stream-native warmup	3,000	-0.018
Four-cond. hardening	800	-0.014

Table 6: Alignment-specific gain across three independent controlled comparisons. Consistent sign across training budgets rules out a single-run artifact.

shorter training budget leaves all conditions closer to initialization. The direction is consistent across independent setups and training scales, providing convergent evidence that the gain is real and not an artifact of a specific configuration.

Gate as a corroborating diagnostic. In the hardened four-condition comparison, the learned gate opens more for aligned (0.177) than shuffled (0.167) or dummy (0.163). The model learns to admit more information from the aligned stream — consistent with it carrying useful content. This corroborates the CE evidence through an independent channel.

These results answer **RQ2 positively**: the alignment-specific effect is small but consistent and corroborated by the gate. It is not explainable by token capacity alone.

6.4 RQ3: S_{plan} Encodes Music-Structural Attributes

RQ3 asks whether S_{plan} embeddings encode music-structural information recoverable by supervised probing. We train linear probing heads on frozen S_{plan} features and compare against a zero-feature baseline — inputs zeroed, with only head bias and BatchNorm statistics available — to expose which axes are explained by dataset priors.

MIR attribute	Aligned S_{plan}	Zero-feature
Instrument macro-F1	0.501	0.426
Instrument micro-F1	0.791	**0.835**
Pitch-class macro-F1	**0.854**	0.830
Meter accuracy	0.783	**0.855**
Weak key accuracy	**0.557**	0.126
Segment density acc.	0.874	**0.879**
Segment polyphony acc.	**0.761**	–

Table 7: MIR probing results. Bold marks the row where the feature-dependent model is more robustly attributable. Label-shuffle and target-prior controls pending.

Axes that support a positive claim. Weak key accuracy (0.557 vs. 0.126, $\Delta = +0.431$) is the strongest signal: the zero-feature baseline is near chance, confirming that the gain is feature-specific. Pitch-class macro-F1 (0.854 vs. 0.830, $\Delta = +0.024$) also shows positive feature-specific signal. Instrument macro-F1 (0.501 vs. 0.426, $\Delta = +0.075$) shows some signal but requires class-imbalance controls before strong claims.

Axes that are prior-dominated and excluded. Instrument micro-F1 (0.791 vs. 0.835, zero-feature *higher*) and meter accuracy (0.783 vs. 0.855, zero-feature *higher*) are largely explained by common-class and common-meter priors. These axes must be excluded from headline claims until label-shuffle controls confirm feature-specific learning.

RQ3 receives a qualified positive answer: S_{plan} encodes key and pitch-class structure well above the prior baseline, while meter and instrument micro-F1 require additional controls before any claim is appropriate.

6.5 Design Constraint: Stream-Native Interface Is Necessary

The positive results in §6.2–§6.4 depend on the stream-native interface design. Two preliminary comparisons demonstrate that naive alternatives fail.

Side-channel insertion. Inserting S_{plan} tokens as additional context rows into the released UniAudio 2.0 interface produced nearly identical results across aligned (5.511), shuffled (5.512), and dummy (5.510) conditions, all worse than reason-only (5.360). The model ignored symbolic content when it was injected as an external side-channel incompatible with multi-stream positional encoding.

LoRA adapter with gating. Adding LoRA adapters stabilized training (valid CE 5.412) but the gate converged near zero (3.36×10^{-4}), indicating the symbolic stream was admitted at negligible magnitude. Training stability is not fusion evidence.

Both results isolate the same root cause: symbolic tokens must enter through the same stream-native positional encoding as R_a and C_a. These failures are not incidental engineering obstacles — they motivated the stream-native design and are reproducible under controlled conditions.

7 Discussion

What we can and cannot claim. The three RQs receive positive or qualified-positive answers under controlled Slakh experiments. The claim boundary is important: (1) all experiments use oracle MIDI input, not predicted symbolic tokens; (2) all CE results are on semantic token prediction, not decoded waveforms; (3) the scope is limited to Slakh synthesized audio. These boundaries are intentional, not evasions — they define what the current evidence supports before the next experimental phase.

Capacity versus alignment specificity. The dominant effect in our primary evaluation is the stream capacity gain ($\Delta \approx -0.245$ CE), and any symbolic stream achieves this. A paper reporting only the comparison to reason-only would overstate the alignment claim. Our multi-condition design separates the two effects, and the alignment-specific effect ($\Delta \approx -0.014$–-0.034 across comparisons) is the meaningful contribution. The gate corroborates it through an independent channel.

Why consistency across scales matters for RQ2. The alignment-specific gap is positive across a proxy task (5,000 steps), a warmup run (3,000 steps), and the hardened four-condition comparison (800 steps, matched conditions). A finding that requires a specific seed or training length is weak evidence; a finding that appears consistently in sign across independent runs at different scales is substantially more robust. The decreasing magnitude is expected: the shorter-budget comparison leaves less room for conditions to diverge.

What MIR probing adds. The probing results connect acoustic utility (RQ1/RQ2) to an interpretable mechanism. The strong weak-key gap ($+0.431$) suggests that S_{plan} captures tonal identity as its most robust representational axis. The prior-dominated meter result is an honest negative: it tells us that 30 frames/window at the current training scale is insufficient to learn meter beyond the common-time prior, and that longer sequences or stronger rhythm supervision may be needed.

8 Limitations

No decoded generation evidence. All experiments evaluate cross-entropy on C_a token prediction under oracle symbolic input. We do not yet have waveform-level evidence that S_{plan} conditioning changes the perceptual quality or structural fidelity of decoded audio. Generating audio from the aligned checkpoint and comparing it against shuffled outputs perceptually remains future work.

Oracle symbolic input. S_{plan} tokens are derived from ground-truth MIDI files. A practical system requires either audio-to-MIDI transcription or a separate audio-to-S_{plan} predictor at inference time, neither of which is trained yet.

No LLM-supervised tokenizer. The current S_{plan} is trained via reconstruction and MIR objectives but not via an LLM decoder objective. A language-model-supervised tokenizer (S_{LLM}) would allow text-language grounding, enabling music captioning and symbolic QA.

Slakh-only scope. All experiments use Slakh, a synthesized paired audio-MIDI corpus. Generalization to real-world recordings requires either transcription or domain adaptation, outside the current scope.

Pending MIR controls. Label-shuffle and target-prior controls for the probing results are incomplete. Instrument macro-F1 and several other MIR axes cannot be claimed as robust symbolic evidence until these controls pass.

9 Conclusion

We introduced S_{plan}, a compact learned symbolic planning token that compresses MIDI-derived musical structure into a fixed-length RVQ stream for integration into unified audio language models. We organized our work around three research questions and found:

- **RQ1 (Acoustic Utility):** Aligned S_{plan} reduces acoustic token cross-entropy in three independent controlled comparisons, most rigorously in our matched four-condition evaluation: test CE 5.600 vs. reason-only 5.859.

- **RQ2 (Alignment Specificity):** Aligned S_{plan} consistently beats both shuffled and dummy controls. The alignment-specific effect ($\Delta \in [-0.034, -0.014]$ CE) is positive and consistent across all three comparisons, corroborated by gate values that open more for aligned content.

- **RQ3 (Music-Structural Content):** MIR probing recovers weak key ($+0.431$ above the zero-feature baseline) and pitch-class ($+0.024$) attributes. Meter and instrument micro-F1 are excluded as prior-dominated.

Two negative results establish stream-native interface as a necessary design constraint, not merely an engineering preference.

The central contribution is methodological as much as empirical: we argue that credible symbolic-fusion claims require multi-condition evaluation with matched corrupted controls (shuffled and dummy), zero-feature MIR baselines, and honest reporting of what each comparison can and cannot support. The same evaluation framework applies broadly to any future work claiming symbolic grounding in audio language models.

Acknowledgments

This work was conducted as part of the CT-NLP course at the Graduate School of Cultural Technology, KAIST.

A Fixed-Window Codec Gallery

Table 8 shows per-window reconstruction statistics for three validation examples from Slakh Track01501.

Window	Binary Acc	Cont. MSE
w000 (0–30 s)	1.000	0.00174
w001 (30–60 s)	1.000	0.00246
w003 (90–120 s)	1.000	0.00130

Table 8: Fixed validation window statistics. All windows achieve perfect binary feature reconstruction. Unique codes per book range from 15–26 across windows, confirming no single-code collapse.

B Per-Window Case Spectrum (Stream-Native Warmup)

Table 9 includes both positive and boundary cases from the stream-native warmup evaluation to prevent cherry-picking the qualitative evidence.

Window	Aligned	Shuffled	Dummy	Margin
w000	5.226	5.301	5.280	+0.054
w001	6.159	6.179	6.163	+0.004
w002	6.097	6.104	6.091	−0.006
w003	5.876	5.893	5.869	−0.007

Table 9: Per-window validation losses. Margin = aligned minus best control; negative values are boundary cases where aligned is not the best.

C Per-Codebook Cross-Entropy (Four-Condition Hardening)

Table 10 shows per-codebook test CE from the four-condition hardening evaluation. The symbolic benefit is distributed across codebooks, with the strongest gains in books 0, 2, and 3.

Condition	cb0	cb1	cb2	cb3	cb4	cb5	cb6	cb7
Reason-only	5.60	6.00	3.45	5.28	6.02	6.48	6.87	7.16
Aligned S_{plan}	**5.24**	5.71	**3.19**	**4.98**	5.75	6.25	6.68	7.01
Shuffled	5.26	5.73	3.20	4.99	5.76	6.26	6.69	7.02
Dummy	5.26	5.73	3.20	4.99	5.76	6.26	6.69	7.02

Table 10: Per-codebook test CE from the four-condition hardening evaluation. Aligned S_{plan} achieves the lowest CE in books 0, 2, and 3.